

CHEM 427/527 · SESSION 1 · THURSDAY, OCTOBER 1, 2026

Data Science for Chemical Measurements

Matt Bush, with teaching assistants Lucas Narisawa and Chris Weir

Drafted with Claude Code. I checked every number on these slides.

FIRST, WHAT YOU NEED TODAY

Three answers before anything else

What you do

Everyone does
worksheet.md: about 60
minutes, no code.

Due on Canvas by 11:59 pm
on Friday, October 2.

Who helps you

Last names A to K: **Lucas
Narisawa**.

Last names L to Z: **Chris Weir**.

Same teaching assistant
through session 3.

What it costs

Sessions 1 to 3 score 3 or 0
for completion. Setup cannot
cost you points.

Working install by **Tuesday,
October 13**.

INSTALL NOT FINISHED?

Start the downloads now, while I talk

1. Open **setup.md** at the step where you stopped.
2. Start the large downloads: VS Code (step 3), uv (step 2), the course environment (step 7).
3. Raise a hand. Your teaching assistant comes to you.

The downloads run in the background, so you lose none of the next 30 minutes.

WHY THE COURSE LOOKS THE WAY IT DOES

Over the next 20 years, I expect that the most successful chemists will have three core competencies:

1

Choose important questions

2

Collaborate with AIs to answer them

3

Check their answers

This course practices all three on chemical measurements. The third is the one you are graded on.

WHERE I COME FROM

Since the early 2000s, code has amplified the impact of my research

- Automating measurements
- Automating analysis
- Numerical estimates of physical observables
- Advanced analysis and statistics, applied in new fields

**5,000 to
10,000 hours**

of traditional software
development, at the interface
with chemical measurements

NOVEMBER 2022 TO 2025

I was a skeptic for a long time

Nov 2022

ChatGPT goes public. Little value next to my domain expertise.

Then

Some value in narrow, limited uses.

Then

Reports of tech companies using AI to automate routine tasks.

What grading taught me

I graded work its authors could not explain, and the gap was easy to see. In this course you explain cells from your own notebook out loud, in the final project check-in.

THE TIPPING POINT, EARLY 2026

An Anthropic co-founder on how Anthropic writes code

"Comfortably the majority of code is being done by the system."

Jack Clark, on The Ezra Klein Show, February 24, 2026

He added that 99% by the end of 2026 was possible, if progress sped up sharply.

My response: an emergency meeting of my research group.

Source: The Ezra Klein Show, "How Fast Will A.I. Agents Rip Through the Economy?", New York Times, Feb 24, 2026

PREDICTIONS, THEN MEASUREMENTS

What has been measured, and by whom

Anthropic, May 2026

Over 80%

of the code merged into Anthropic's own codebase was written by Claude. Measured by the company that sells the tool.

METR, independent

Early 2025: 16 experienced developers, 246 tasks. 19% slower with AI, while believing they were 20% faster.

Feb 2026: METR calls that result outdated. Developers now avoid tasks they would have to do without AI.

The gap between what people believed and what was measured is a measurement problem. Ask what was measured, and how.

Sources: Anthropic, "When AI Builds Itself" (2026); METR, study of early-2025 AI (Jul 10, 2025) and design update (Feb 24, 2026)

WHAT CHANGED, PART 1

From a chat bot to an agent

Chat bot

The AI most of you have used

- You paste text into a window
- It answers in text
- You copy code out and run it
- You paste the error back in

Agent

Claude Code, in VS Code, from today

- It reads the files in your project
- It runs the code itself
- It reads the error and fixes it
- It reports what it changed

Three advances in the models themselves

Cross-domain expertise

Chemistry, physics, electrical engineering, biology, statistics and code, in one conversation. My own work has drawn on all of them.

Alignment

It does what you asked, not what it guessed you meant. Ideally, it says when it cannot or is unsure.

Persistence

It stays on one task for minutes to hours, and ideally beyond, without losing the thread.

MY EVIDENCE

A high-speed digitizer with on-board processing

18 months

A trainee and I, with traditional practices. No headway bringing it into our measurements.

< 1 month

With an AI-assisted workflow, I brought it online.

< 1 week

For the team to go from resisting software and hardware changes to not turning back.

WHY IT WORKED

Better models, an agent, and practices I had to learn

The practice that made it work	Where you practice it	Session
Verification	Give it a check it can run	6
Explore, then plan, then code	Plan a figure before any code	7
Context management	Correct early and clear context	8

I learned these in workshops and in discussions with software engineers. The course teaches one practice per session from session 4 on.

HOW I CODE NOW

I write more code now than at any point in my career. I type almost none of it by hand.

My time goes to deciding what to build, reading what the AI wrote, and testing it against the instrument.

MY GREATEST CONCERN

AI raised homework scores and lowered exam scores

+18%

homework scores

−30%

homework completion time

−20%

monthly exam scores, within 6 months

−18 to

−24%

entrance exam scores, full effect after about 2 years

About 80% of the students using AI showed signs of outsourcing their homework: very short completion times with high scores. The harm came from how the tool was used.

26,811 students in grades 7 to 12 in China, followed for 30 months as schools adopted AI at different times. An observational study, not a randomized trial.

Source: Strömberg, Lei and Wu, "The Generative AI Learning Penalty," CEPR Discussion Paper 21577, June 2026

MY PLEA: BE CURIOUS, AND PUT YOUR LEARNING FIRST

Use the AI to supplement your understanding

- Ask why it wrote a line the way it did, and what it would do differently.
- Ask technical questions, and ask for answers aimed at what you already know.
- Try a step yourself first, then compare with its version.
- Ask it to quiz you on what it just did.
- Check every result against your own data.

A prompt to try

Explain this cell to a chemist who knows Beer-Lambert but has never used pandas.

My own practice: I ask technical questions and ask for answers aimed at my specific expertise.

HOW THE COURSE TEACHES IT

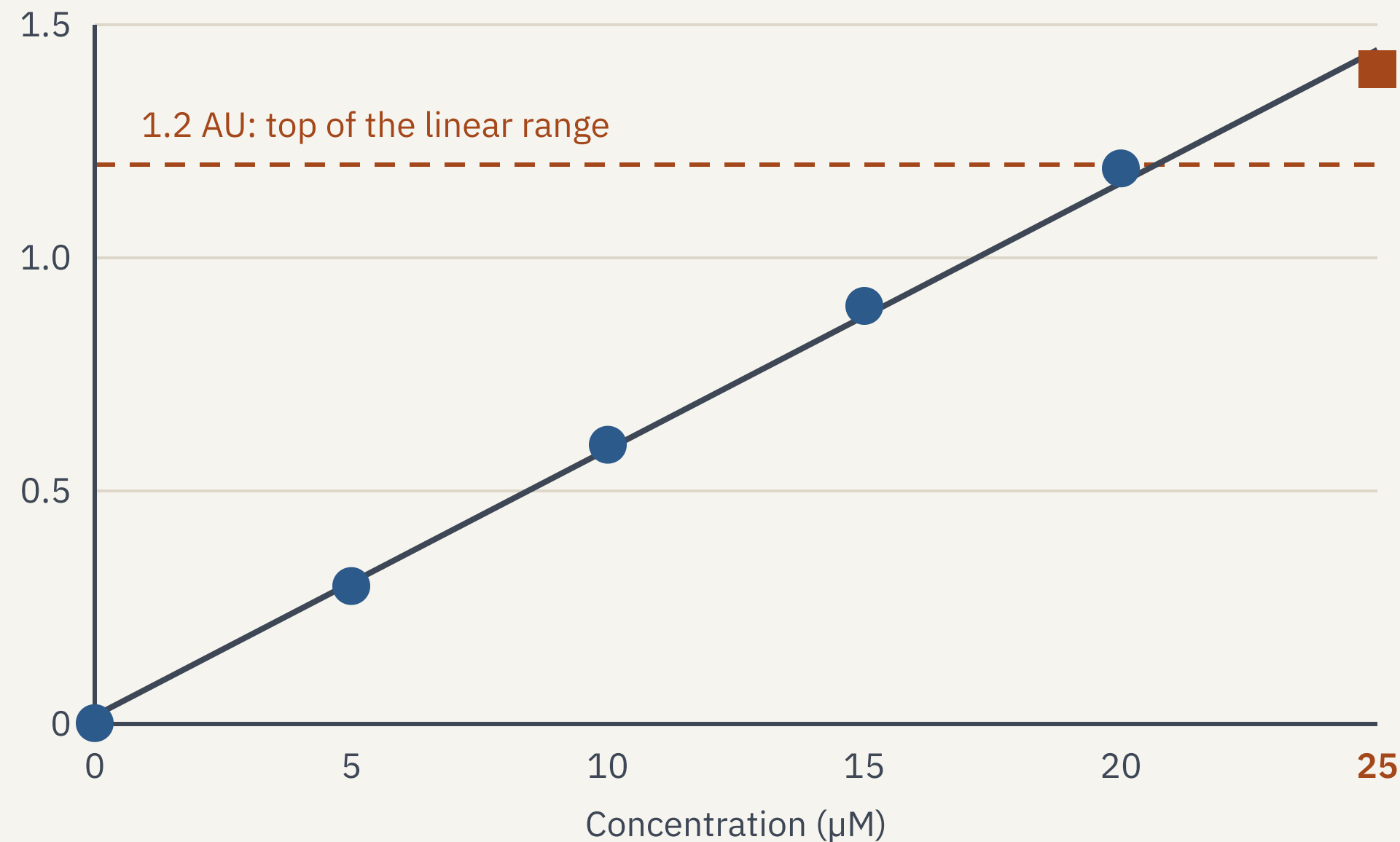
One practice per session, then used in your own work

Session	Practice
4	Ask questions as you would a senior colleague
5	Specific prompts: name the file, the column, the constraint
6	Give it a check it can run
7	Explore and plan before code
8	Correct early and clear context
10	Edit the CLAUDE.md file
12	A skill file for a step you repeat
15	A fresh-context review before you submit

WHERE THE CODE ENDS

Six standards, one straight line, $R^2 = 0.997$

Absorbance (AU)



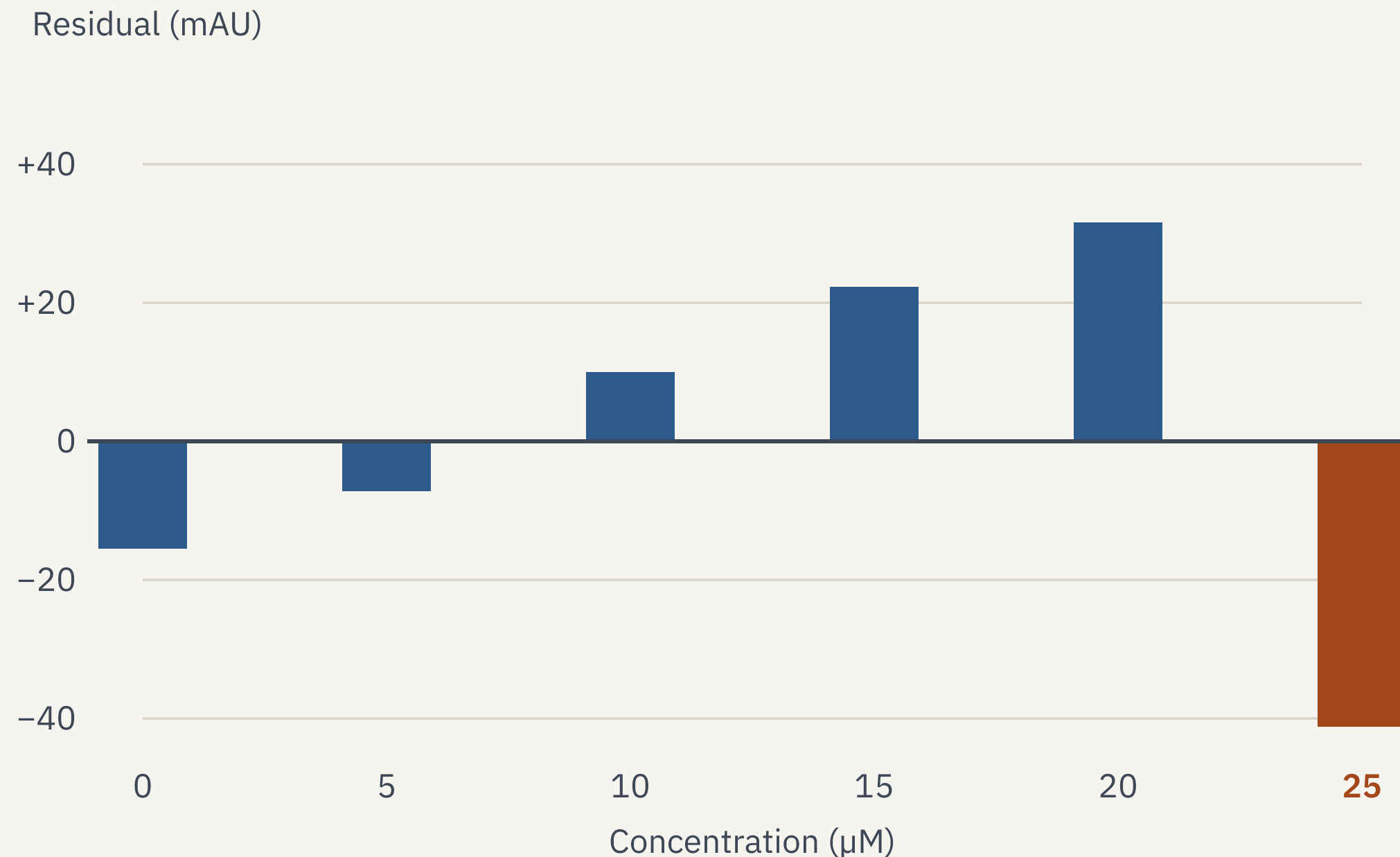
A spectrophotometer at 520 nm, 1 cm cell. The line through all six gives a slope of **0.0571 AU/μM**.

Circles: the five standards below 1.2 AU.
Square: 25 μM, the only one above.

Writing this fit takes one line of Python.
The assistant writes it in seconds.

WHERE THE CHEMISTRY BEGINS

The residuals say the line is wrong



They rise across the first five standards, then drop at 25 μM, the only standard above 1.2 AU.

Refit without it: slope **0.0596 AU/μM**, 4.3% higher. Every concentration read from the line falls by about 4%.

The line of Python is free. The decision about 25 μM is the course, and it is what you are graded on.

HOW THE COURSE RUNS

The analysis from your chemistry courses, on full instrument files

What	How
Data	Chromatography, mass spectrometry, spectroscopy, imaging
Tools	Python every session, Claude Code from today
Grading	Your decisions, in the judgment cell, from session 4
Final project	Topic Nov 8, draft Nov 22, final Dec 9, presentations Dec 10

EVERY TUESDAY AND THURSDAY

How a session runs

20 min

Opening: the measurement and the method. Today runs longer.

~100 min

Activity: you work in **activity.ipynb** while we circulate.

3rd hour

Office hour: stay to finish or to ask.

Run **git pull** before every class. From session 4, each notebook scores 0 to 3: 1 point for running top to bottom after a kernel restart, 0 to 2 for the judgment cell.

The rules that decide how your work is graded

1. You own every cell

In the final project you explain cells from your own notebook out loud.

2. Explanation before code

Type "Explain what this cell does" before "fix this".

3. Verify against your data

Check the units, the magnitudes and the plot.

4. One judgment entry

One suggestion you changed, rejected or verified, with a reason from your numbers.

CLAUDE.md asks the assistant to explain before it writes and to leave the judgment cell empty. It guides and enforces nothing, and you can edit it. **Privacy:** set the Model Improvement switch the way you want. **Limits:** Pro usage is shared between claude.ai and Claude Code.

YOUR SHORT-TERM GOAL

The tools on your laptop by October 13

Tool	What it does	setup.md
git	Downloads the course, then updates it with git pull	1, 6
uv	Installs Python 3.14 and the course environment	2, 7
VS Code	Where you run notebooks and work with Claude Code	3
Claude Pro	The account Claude Code needs: \$20 a month, about \$60 for the quarter	4
Claude Code	The agent, inside VS Code	5
The course repository	Lives on GitHub. You do not need a GitHub account	6

You are done when **setup-check.ipynb** prints **Setup check passed**.

IF YOUR INSTALL IS NOT WORKING YET

Four steps to a working laptop

Today

Finish with your teaching assistant.

S2 and S3

Pair with a working laptop. You type. One notebook, both names.

End of S3

Checkpoint. Still failing: a named office-hour appointment.

Tue Oct 13

Deadline. Every notebook on your own laptop.

When something fails, paste the full error text into the setup status survey on Canvas. The exact text is how we find the fix.

THE NEXT 85 MINUTES

Today's two tracks

Install works

Run git pull, then **activity.ipynb**.
Type its two prompts. Check each
explanation against what the cell
printed.

Install not working

Go to your teaching assistant now.
Do the worksheet while something
installs.

Everyone

worksheet.md, no code, on Canvas
by Friday 11:59 pm.

Choose important questions. Collaborate with AIs to answer them. Check their answers. Today, the check is the worksheet.